



Society of Petroleum Engineers

SPE-230381-MS

Next-Gen Fracture Diagnostics: AI-Decoded Water Hammer Physics for Adaptive Design and Field-Wide Optimization

R. M. Iuzofatov, SPE; I. A. Seleznev, A. A. Borisenko, and A. V. Fedorov, STISS LLC, Russia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230381-MS](https://doi.org/10.2118/230381-MS)

This paper was prepared for presentation at the SPE Caspian Technical Conference and Exhibition scheduled to be held in Baku, Azerbaijan, 25 – 27 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Hydraulic fracturing is performed to intensify hydrocarbon production by applying high pressure to the productive formation in order to initiate the creation of a fracture and its further development in the formation with subsequent placement of a propping agent (proppant) to ensure inflow to the production well. The job is performed as per preliminary design according to the model, which is necessary for achieving the optimal parameters of the fracture. At the modeling stage, the expected parameters of the hydraulic fracture and production rate assessment is made on the basis of information on the physical properties of the formation according to core studies and well logging data, data from previously performed fracturing jobs on adjacent wells, a job plan, information on the types and properties of fracturing fluids and proppants.

After conducting hydraulic fracturing and calibrating the fracture model based on the obtained data, it becomes possible to form a more accurate representation of the actual fracture parameters for productivity evaluation. However, this representation comes with a significant degree of uncertainty regarding the actual fracture geometry and fracture planes propagation direction, the uniformity of change in the fracture half-length within the reservoir in both directions from the wellbore (symmetry), and the distribution of proppant volume between the fracture planes. Under these circumstances, making decisions regarding the optimization of the hydraulic fracturing program for a field remains a highly challenging task.

For the studied waterhammer signals, it is difficult to obtain the true values of the fracture parameters; the available data is a result of related inverse problems' solving.

Recording the pressure signal with a high sampling rate (in practice, up to 200 Hz) makes possible a detailed analysis of the oscillations observed after pumps shutdown (Ma et al., 2019; Qiu et al., 2023). One of the problems that can be solved by analyzing the high-frequency pressure signal is the localization of the hydraulic fracturing zone (Oparin et al., 2024). Knowing the fracturing zone and assuming the influence of the fracture geometry on the shape of the recorded signal, the task of estimating the fracture parameters is set.

The waterhammer signal recorded at the wellhead can have a complex shape due to the well design, fluid properties, fracture geometry, and reservoir properties. The variable velocity of acoustic wave propagation along the wellbore further complicates the solution of the inverse problem.

Currently, there is an increasing number of signal analysis techniques based on the utilization of various novel neural network algorithms. The development and implementation of these solutions in production

processes allows to explore new approaches to data management and significantly reduce production costs. On the other hand, it is of the highest priority for the oil industry to ensure the reliable operation and safety of any forecasts, and therefore, it is necessary to monitor the physical plausibility of neural network algorithms or apply it in combination with robust mathematical modeling.

This paper proposes a method for estimating fracture parameters based on the analysis of waterhammer signals. For this purpose, the use of the latent space parameters of a variational autoencoder trained on a set of simulated waterhammer data is proposed. Based on the field data analysis results, the possibility of additional estimation of fracture parameters is demonstrated.

Methodology

The described method consists of two main parts: direct modeling of waterhammer signals using the Hydraulic Impedance Testing (HIT) method (Paige et al., 1995), and the use of a neural network based on a variational autoencoder (VAE) scheme (Doersch, 2016) which is trained on simulated data, to estimate changes in the model parameters of the signal.

HIT method

For modeling the training dataset of waterhammers, a model using the acoustic approximation is applied. The solution is performed in the frequency domain, transition to which is carried out using the Laplace transform. The solution is represented as:

$$p(x, s) = P_{\rightarrow} e^{-\gamma(s)x} + P_{\leftarrow} e^{\gamma(s)x} \quad v(x, s) = V_{\rightarrow} e^{-\gamma(s)x} + V_{\leftarrow} e^{\gamma(s)x},$$

where $s = \sigma + i\omega$ – is the Laplace transform variable. The following also holds true:

$$P_{\rightarrow}(s) = Z_c(s) Q_{\rightarrow}(s) \quad P_{\leftarrow}(s) = -Z_c(s) Q_{\leftarrow}(s),$$

where $Z_c(s) = \frac{\rho c}{s}$ – is the characteristic impedance of the segment. For a segment with a known characteristic impedance, a system of equations for transferring the hydraulic impedance $Z(x, s)$ from one end of the segment to the opposite end is formulated:

$$Z(x_1, s) = Z_c(s) \frac{1+R(x_1, s)}{1-R(x_1, s)} = Z_c(s) \frac{1+e^{2\gamma(s)(x_1-x_2)} R(x_2, s)}{1-e^{2\gamma(s)(x_1-x_2)} R(x_2, s)}$$

$$R(x_2, s) = \frac{Z(x_2, s) - Z_c(s)}{Z(x_2, s) + Z_c(s)}.$$

Thus, the sequence of characteristic impedances of the segments defines the physical configuration of the system. To account for the influence of the fracture, the formula for the parallel connection of segments is used at the point where the fracture adjoins the wellbore:

$$p_1 = p_2 = \dots = p_n$$

$$\frac{1}{Z_1} = \frac{1}{Z_2} + \dots + \frac{1}{Z_n} + \frac{1}{\zeta(s)},$$

Where $\zeta(s) = R + \frac{1}{sC} + sL$ – is the lumped impedance, which depends on the fracture parameters and affects the signal propagation along the wellbore. After setting the no-flow boundary condition at the bottom of the well, the impedance transfer equation is used for each allocated segment until the wellhead is reached. A second boundary condition is set at the wellhead in the form of a given flow rate function 'Q', which satisfies the requirements for performing the inverse Laplace transform and describes the fluid injection process during hydraulic fracturing. The final solution in the time domain is obtained after calculating the inverse Laplace transform at the wellhead.

Procedural modeling was used to create the training dataset of waterhammers, varying several system parameters: geometric parameters of the fracture model, fracture depth, fracture fluid parameters, wellbore

fluid parameters, and source function parameters. Furthermore, several wellbore configurations were used in the modeling to reflect the influence of various pipe joints on the resulting signal. Fig.1 shows examples of simulated waterhammers used for training.

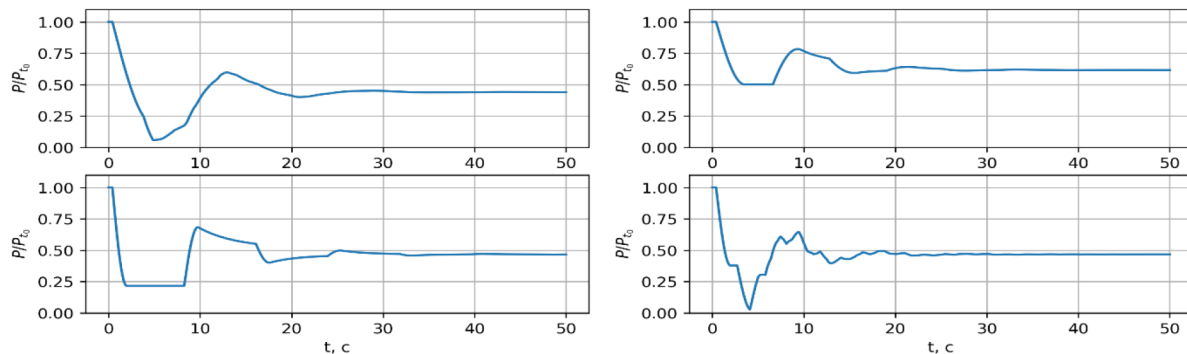


Figure 1—Examples of training dataset signals.

VAE scheme

A neural network based on a variational autoencoder scheme is used for training on model signals. It is crucial to mention that learning is unsupervised. In other words, during training this neural network does not use the model parameters of the signal as prediction targets but learns to reconstruct the signal based on a chosen metric. By learning to extract core features from the input signal and decoding it to latent representation, the neural network forms latent space distributions. This allows the use of the latent space for working with signal parameters.

Using the encoder, each real signal is uniquely mapped to a point in the latent space. Since the latent space vector is the carrier of all available information for further decoding, vectors of similar signals will be close. Analyzing the parameters of the nearest neighbors allows obtaining the parameters of the model signals that are most similar to the input unknown signal from the perspective of the chosen metric. The averaged parameter values of the neighbors, or the selection of parameters using any clustering method, can be used as an initial approximation for an accurate initial approximation of the inverse problem (Petrov et al., 2023) or for estimating the model parameters of the fracture. However, for such an analysis of a single signal, a preliminary analysis of the inverse problem is necessary. For example, if there are dependent parameters in the problem, and their mutual change can be compensated, then a clear distribution in the latent space cannot be achieved, since in this case, changes in the parameters of the initial model do not affect the final signal.

Since this model does not account for the spatial characteristics of the fracture but represents it as a parallel segment of the well, the solution of the inverse problem based on a single signal contains significant assumptions. In order to benefit from the resulting distributions in the latent space, instead of analyzing single signal we propose finding the difference between successive responses to estimate the evolution over time during the work process. In this case, all system parameters for both signals remain unchanged and any differences between them are attributed solely to variations in the model's fracture parameters. To assess the quality of the performed hydraulic fracture, one can study the dynamics of the change in the fracture response signal, for example, by comparing datafrac (or mini frac) waterhammers with main fracturing job waterhammers within the same stage.

Despite the inaccurate coincidence of the reconstructed signal with the original field signal, the signal sequence retains its key differences. The changes in the latent space are of particular importance, and the analysis of these shifts is of primary interest.

In the context of using the autoencoder, each signal has a uniquely corresponding point in the encoded latent space, which can be used to investigate the relative characteristics of two signals: to extract the vector responsible for increasing one of the system parameters. Using model data, we subtract the averaged latent

representation of signals with a small parameter value from the averaged latent representation of signals with a large parameter. Thus, we obtain an abstract vector in the latent space, which corresponds to the averaged direction of increase of the selected parameter. An abstract increase vector is unique for each single parameter in mathematical modeling. We analyze only those parameters where the projection of the latent representation increases monotonically with an increase in the model parameter. In this case, the parameters are the components of fracture impedance. If the monotony is violated, it indicates uncertainty in the distribution in latent space and makes it unreasonable to base further analysis on the average representation. An example of such a parameter would be the water hammer celerity. Fig. 2 shows a graph of the monotonicity of parameters in the latent space. To construct it, an increase vector is allocated, on which the difference between the successively increasing parameters is sequentially projected. As can be seen on the left plot of Fig. 2, the latent representations of fracture's half-length are distributed monotonically, thus they could be analyzed by the proposed method. On the other hand, right plot of Fig. 2 shows distribution of water hammer celerity and demonstrates the uncertainty in determining the change using a scalar product.

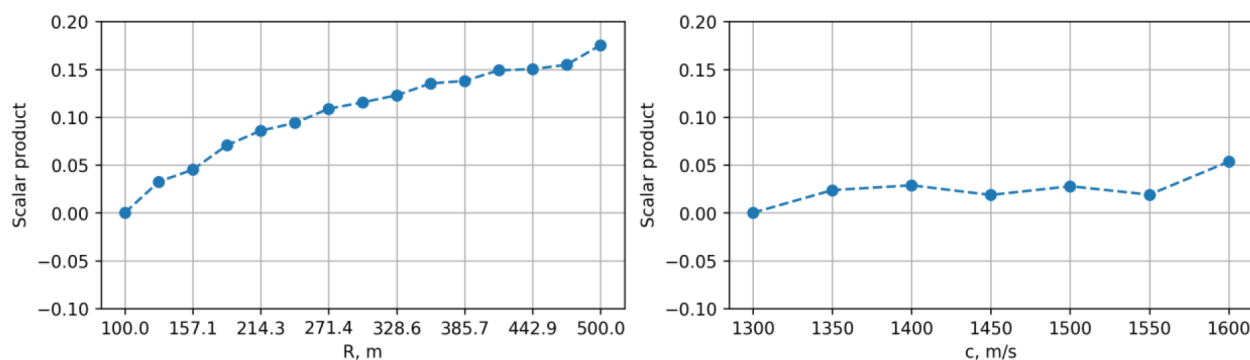


Figure 2—Example of projections of the distribution of an increasing model parameter onto the corresponding increase vector in the latent space.

On the contrary, we may observe the inherent physicality of the obtained result of non-monotonous distribution of certain parameters. For instance, if the water hammer celerity was unambiguously determined by a solitary signal, it would suggest an artefact in the modelling process, allowing for non-physical inferences.

Then, for two field signals, we obtain the vector difference of their latent representations, project it onto the pre-calculated parameter increase vector, and based on the projection value, we estimate the change in the model parameters of the system. For example, if the vector difference of two signals from the same stage, calculated in this way, is co-directed with the abstract vector of fracture width increase, it is concluded that one of the results of the operations performed was an increase in the model parameter of fracture width.

Case History

Example 1

The described methodology applied to the field data is presented in Fig. 3. The dataset under analysis comprises a number of water hammer events – an injection test, a mini-frac, and a main frac for one stage. In this case, all other variables in the system are held constant, including the well configuration, reflection from joints, and water hammer celerity for these events.

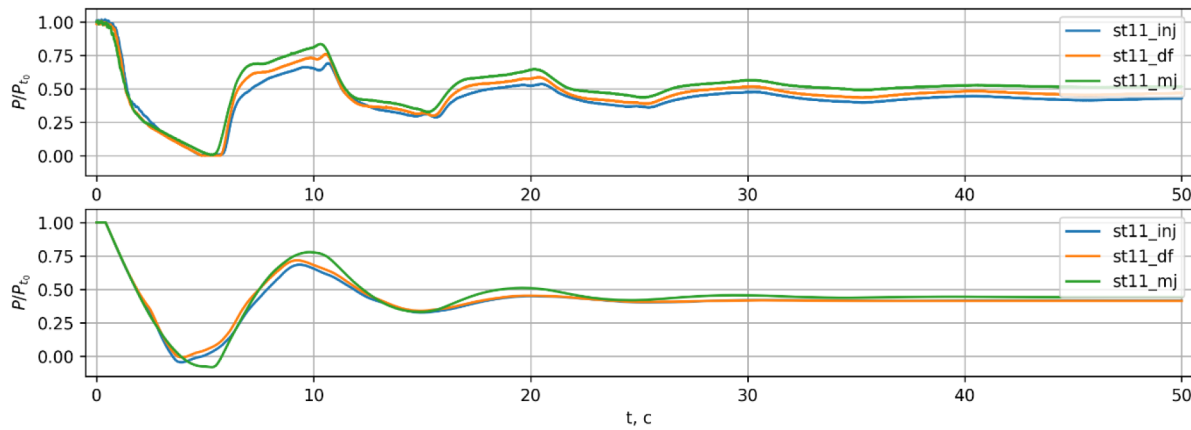


Figure 3—Example of 3 signals from one stage. The corresponding decoded waterhammers are shown on the lower graph.

During processing, we obtain 2 vectors of parameter changes in the latent space, as described above. To illustrate the relative change, we take the initial parameter values of the first signal as 1, then subsequent parameter changes are calculated relative to the first one using the formula:

$$x_{n+1} = x_n \left(1 + \frac{\overrightarrow{\Delta X} \cdot \overrightarrow{f_2 - f_1}}{\|\overrightarrow{\Delta X}\| \|\overrightarrow{f_2 - f_1}\|} \right)$$

where $\overrightarrow{\Delta X}$ – is the pre-calculated parameter change vector, and f_1, f_2 – are the latent representations of the corresponding field signals.

Fig. 4 shows the result of the signal analysis. The water hammer signals in Fig. 4 are normalized to the maximum values for clarity of presentation. In this case, one can see a consistent increase in the amplitude of the signal reflection, which indicates an increase in the fracture parameters as a result of hydraulic fracturing operations. Through an assessment using the latent space, it is possible to determine the impact of the work performed immediately after pressure release in the well. The graph on the right picture of Fig. 4 shows a sequential increase in the model fracture parameter as a result of the hydraulic fracturing operation.

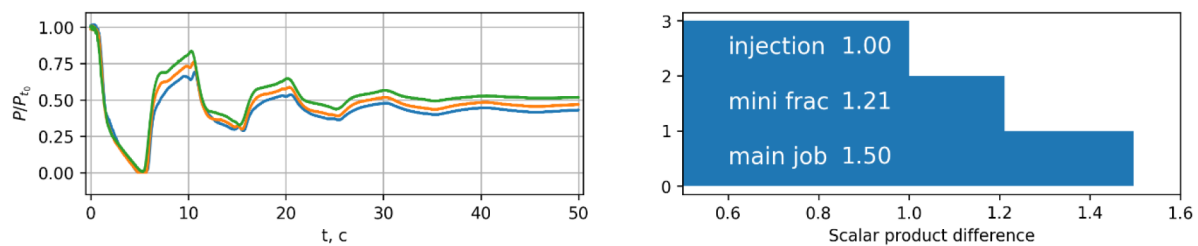


Figure 4—Example of the algorithm's operation on field data from one stage.

Example 2

To demonstrate the operation of the method, we will compare estimates from the proposed method with reported calculated metrics for increasing fracture opening width. At each stage of this dataset, there are mini-frac and main-frac jobs an analysis of the difference between which is shown in the Fig. 5.

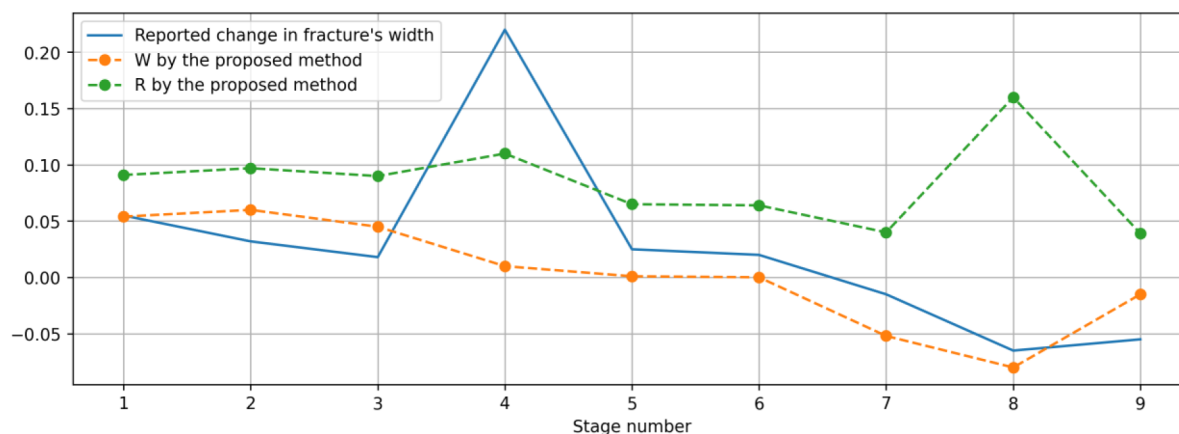


Figure 5—Example of the algorithm's operation on field data from several stages.

The blue line of the graph shows the reported difference in fracture width before and after hydraulic fracturing operations while the dotted lines show estimates of the change in width and half-length using the proposed algorithm at each stage. It is important to mention a consistent polarity of changes in the reported parameters and the estimated values provided by the method. Since the fracture impedance of the model depends on a combination of width and half-length parameters, the detailed numerical representation depends on the joint calculation of changes in the estimation parameters.

An important result is the speed of obtaining estimates of the effectiveness of the use of intensification methods. While detailed estimates based on adjusting well parameters through software modeling require time for data collection and the work of a model expert, estimates using the proposed method involve analyzing effectiveness immediately after completion in the automatic background mode. Proposed approach allows for the early detection of potential faults related to hydraulic fracturing operations, and enables the adjustment of the work plan on-site.

Conclusions

Neural networks are increasingly playing a significant role in digital data processing algorithms. By leveraging existing architectures and customizing them to meet specific needs, experts in the field can greatly enhance their day-to-day operations. At the same time, it is crucial in the oil industry to minimize errors associated with the statistical nature of neural networks in order to prevent inaccurate forecasts. To achieve this, it may be possible to utilize physically informed neural networks or develop one's own robust algorithms designed to clarify the solution to inverse problems or accelerate optimization techniques.

This paper presents a novel approach to using the latent space of a generative neural network trained on the generated dataset of water hammer events to assess changes in fracture parameters during hydraulic fracturing operations. In combination with the monotonicity of the projection of model parameter changes in the latent space, as illustrated in Fig. 3, this approach may be considered physically plausible. The automation of routine tasks through the implementation of advanced algorithms can significantly reduce costs and allow for the use of innovative tools for quality assurance and prevention of work-related errors.

Based on the results of this work, an approach to solving the problem of additional fracture parameter estimation during hydraulic fracturing by analyzing the waterhammer event is described:

1. A novel methodology that combines physics-based modeling (HIT) with machine learning (variational autoencoder) to interpret complex waterhammer signals for hydraulic fracture characterization, moving beyond traditional inverse problem techniques is proposed.
2. The research validates that the latent space of a VAE, trained solely on signal reconstruction, effectively encodes meaningful information about the physical parameters of the fracture system. This

allows for the estimation of fracture properties by analyzing the proximity and vector relationships between data points in this compressed representation.

3. The method is particularly powerful for conducting comparative analysis. It can reliably track relative changes in fracture parameters (e.g., an increase in width or length) between different stages (e.g., mini frac vs. main frac) on the same well, providing valuable insights into the fracture's evolution.
4. The technique offers a significant practical advantage by generating a well-informed initial approximation of fracture parameters. This can drastically improve the convergence speed and accuracy of subsequent, more precise, inverse problem-solving methods.
5. The conclusion rightly acknowledges the model's simplifications (e.g., representing the fracture as a lumped impedance) and the significant uncertainties inherent in fracture diagnostics. The method is positioned as a supplementary tool for evaluation rather than a definitive standalone solution, highlighting a scientifically honest approach.

Further development of the methodology for improving the forecast may involve both a more in-depth analysis of the physics of the phenomenon under study and the addition of various signals of different nature to the shared latent space. In the latter case, the existence of a shared latent space would allow the integration of solutions from various mathematical models into a unified view of the developing production facility, which would lead to a more in-depth analysis of events and improvement of existing forecasts.

References

- Doersch, C., 2016, Tutorial on variational autoencoders: arXiv preprint, arXiv:1606.05908.
- Ma, X., Zhou, F., Ortega Andrade, J. A., Gosavi, S., and Burch, D. 2019. Evaluation of Water Hammer Analysis as a Diagnostic Tool for Hydraulic Fracturing. Presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Denver, Colorado, USA, 22-24 July. URTEC-2019-935-MS. doi: <https://doi.org/10.15530/urtec-2019-935>.
- Oparin, M., Crawford, E., Alzahrani, B., Tineo, R., Najmi, A., Enkababian, P., Borisenko, A., and Fedorov, A. 2024. Nonintrusive Rapid Evaluation of Fracture Isolation in Plug-And-Perf Completions Based on Data Science Methodology. Presented at the International Petroleum Technology Conference, Dhahran, Saudi Arabia, 12-14 February. IPTC-24435-EA. doi: <https://doi.org/10.2523/IPTC-24435-EA>.
- Paige, R. W., Murray, L. R., and Roberts, J. D. M. 1995. Field Application of Hydraulic Impedance Testing for Fracture Measurement. *SPE Prod & Fac* **10** (1): 7–12. SPE-26525-PA. <https://doi.org/10.2118/26525-PA>.
- Petrov, I.B., Stankevich, A.S., Vasyukov, A.V. 2023. On Obtaining Initial Approximation For Full Wave Inversion Problem Using Convolutional Neural Network (in Russian). *Russ. Akademija Nauk Mat. Inform. Processy Upr.* **512** (1): 58–64.
- Qiu, Y., Hu, X., Zhou, F., Yi, P., Wang, Y., and Jiang, Z. 2023. A New Investigation of Wellbore Parameters Influence on Shut-In Pressure Transient Signatures from Water Hammer Fracture Diagnostics. Presented at the SPE International Hydraulic Fracturing Technology Conference and Exhibition, Muscat, Sultanate of Oman, 12-14 September. SPE-215709-MS. doi: <https://doi.org/10.2118/215709-MS>.